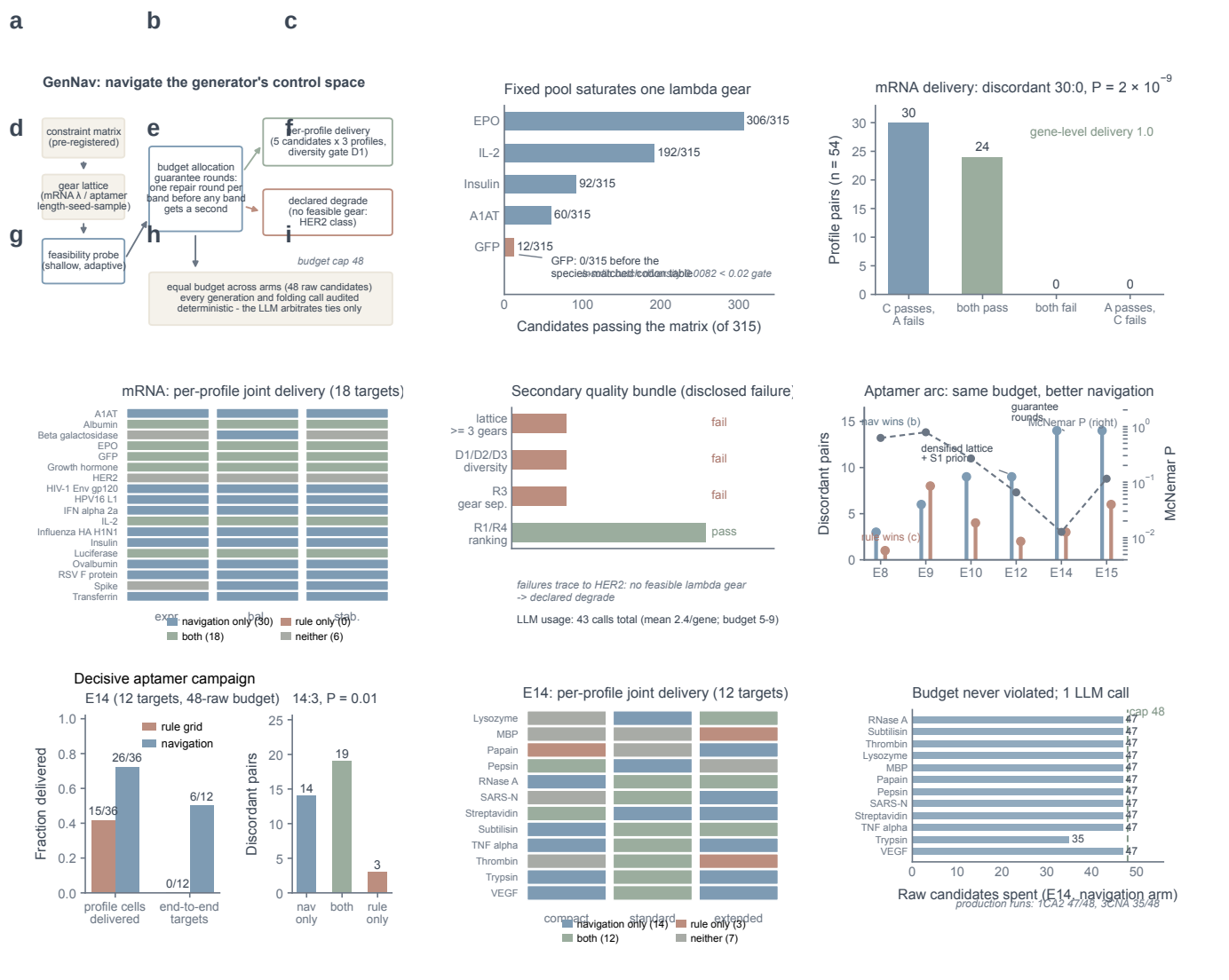

a
b
c
GenNav: navigate the generator's control space
d
constraint matrix (pre-registered)
gear lattice (mRNA λ / aptamer length-seed-sample)
feasibility probe (shallow, adaptive)
e
budget allocation guarantee rounds: one repair round per band before any band gets a second
f
per-profile delivery (5 candidates x 3 profiles, diversity gate D1)
declared degrade (no feasible gear: HER2 class)
g
h
i
budget cap 48
equal budget across arms (48 raw candidates) every generation and folding call audited deterministic - the LLM arbitrates ties only
Fixed pool saturates one lambda gear
EPO
IL-2
Insulin
A1AT
GFP
306/315
192/315
92/315
60/315
12/315
GFP: 0/315 before the
Candidates passing the matrix (of 315)
mRNA delivery: discordant 30:0, P = 2 × 10−9
Profile pairs (n = 54)
30
24
0
0
gene-level delivery 1.0
C passes, A fails
both pass
both fail
A passes, C fails
mRNA: per-profile joint delivery (18 targets)
A1AT
Albumin
Beta galactosidase
EPO
GFP
Growth hormone
HER2
HIV-1 Env gp120
HPV16 L1
IFN alpha 2a
IL-2
Influenza HA H1N1
Insulin
Luciferase
Ovalbumin
RSV F protein
Spike
Transferrin
expr.
bal.
stab.
navigation only (30)
rule only (0)
both (18)
neither (6)
Secondary quality bundle (disclosed failure)
lattice >= 3 gears
D1/D2/D3 diversity
R3 gear sep.
R1/R4 ranking
fail
fail
fail
pass
failures trace to HER2: no feasible lambda gear
-> declared degrade
LLM usage: 43 calls total (mean 2.4/gene; budget 5-9)
Aptamer arc: same budget, better navigation
Discordant pairs
McNemar P
nav wins (b)
rule wins (c)
densified lattice + S1 prior
guarantee rounds
McNemar P (right)
E8
E9
E10
E12
E14
E15
Decisive aptamer campaign
E14 (12 targets, 48-raw budget)
14:3, P = 0.01
rule grid
navigation
Fraction delivered
15/36
26/36
0/12
6/12
profile cells delivered
end-to-end targets
Discordant pairs
14
19
3
nav only
both
rule only
E14: per-profile joint delivery (12 targets)
Lysozyme
MBP
Papain
Pepsin
RNase A
SARS-N
Streptavidin
Subtilisin
TNF alpha
Thrombin
Trypsin
VEGF
compact
standard
extended
navigation only (14)
rule only (3)
both (12)
neither (7)
Budget never violated; 1 LLM call
cap 48
RNase A
Subtilisin
Thrombin
Lysozyme
MBP
Papain
Pepsin
SARS-N
Streptavidin
TNF alpha
Trypsin
VEGF
47
35
Raw candidates spent (E14, navigation arm)
production runs: TCA2 47/48, SCNA 35/48